

1 Four Ways to Approximate a Sampling Distribution

To make inferences we typically need to know the distribution of a statistic $Y = g(\mathcal{X}) = g(X_1, \dots, X_n)$, i.e., its sampling distribution. There are four main strategies for calculating or approximating a sampling distribution:

1. Exact calculation: In rare cases, typically under strong parametric assumptions about the distribution of the data (e.g., a random sample from a normal population), we can calculate the finite-sample distribution of Y exactly.
2. Monte Carlo simulation: If we know (or postulate) the distribution of the data, we can simulate many artificial samples from it, calculate the statistic in each one, and use the simulated draws to approximate the sampling distribution. Here we control the accuracy of the approximation: larger numbers of simulations lead to better accuracy, bounded only by the power of our computers.
3. Asymptotic approximation: If the sample size is large, the sampling distribution may be well-approximated by its limit, relying on the law of large numbers, the central limit theorem, the delta method, Slutsky's theorem, and the like. This requires no knowledge of the distribution of the data beyond, say, moment conditions, but we do not control the quality of the approximation in any given sample.
4. The bootstrap: Approximate the unknown population distribution itself by an estimate of it, and then proceed as in a Monte Carlo simulation, simulating from the estimated distribution as if it were the truth.

Note 2 developed the first three strategies. This note supplies the fourth. The key ingredient is an accurate estimate of the population cumulative distribution function: we first show that the empirical distribution function is such an estimate, and then use it to construct the bootstrap.

2 The Empirical Distribution Function

If we have a random sample $X_1, \dots, X_n$ of size n , then the empirical distribution function $\hat{F}_n$ is the cdf of the distribution that places mass $1/n$ at each data point X_i . Thus, by definition,

$$\hat{F}_n(x) = \sum_{i=1}^n I\{X_i \leq x\}/n,$$

where $I\{\cdot\}$ stands for the indicator function, i.e. the function which equals 1 if the statement in brackets is true, and 0 otherwise. In other words, $\hat{F}_n(x)$ shows the fraction of observations with a

value smaller than or equal to x . An important property of an empirical distribution function is given next.

Theorem 1. *If we have a random sample $X_1, \dots, X_n$ of size n from a distribution with cdf F , then for any $x \in \mathbb{R}$, $\mathbb{E}[\hat{F}_n(x)] = F(x)$ and $V(\hat{F}_n(x)) = \frac{F(x)(1-F(x))}{n} \rightarrow 0$ as $n \rightarrow \infty$. As a consequence, for any $x \in \mathbb{R}$,*

$$\hat{F}_n(x) \rightarrow_p F(x)$$

as $n \rightarrow \infty$.

Proof. Note that $I\{X_i \leq x\}$ equals 1 with probability $P\{X \leq x\}$ and 0 otherwise. Thus, $\mathbb{E}[I\{X_i \leq x\}] = P\{X \leq x\} = F(x)$. Hence $\mathbb{E}[\hat{F}_n(x)] = F(x)$ by linearity of expectation. In addition, $V(I\{X_i \leq x\}) = F(x)(1 - F(x))$ by the formula for variance of a Bernoulli ($F(x)$) distribution. Therefore,

$$V(\hat{F}_n(x)) = \sum_{i=1}^n V(I\{X_i \leq x\})/n^2 = F(x)(1 - F(x))/n \rightarrow 0.$$

To prove the second part of the theorem, we have $\mathbb{E}[(\hat{F}_n(x) - F(x))^2] = V(\hat{F}_n(x)) \rightarrow 0$ as $n \rightarrow \infty$ since $\mathbb{E}[\hat{F}_n(x)] = F(x)$. Convergence in probability then follows from Chebyshev's inequality. $\square$

Actually, much stronger results hold that establish (i) the empirical distribution function converges to the cdf uniformly in x , and (ii) it does so at a “fast rate.”

Theorem 2.

1. Glivenko-Cantelli: If $X_1, \dots, X_n$ is a random sample from a distribution with cdf F , then

$$\sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| \rightarrow_p 0.$$

2. Dvoretzky-Kiefer-Wolfowitz (DKW) Inequality: If $X_1, \dots, X_n$ is a random sample from a distribution with cdf F , then for any $\epsilon > 0$

$$P(\sup_x |F(x) - \hat{F}_n(x)| > \epsilon) \leq 2 \exp(-2n\epsilon^2).$$

The DKW inequality is the uniform analogue of Hoeffding's inequality from Lecture Note 3: at each fixed x , $\hat{F}_n(x)$ is a sample average of the bounded random variables $I\{X_i \leq x\}$, so Hoeffding's inequality already delivers the bound $2 \exp(-2n\epsilon^2)$ pointwise. The remarkable content of DKW is that the same bound holds for the supremum over all x simultaneously.

3 Introducing the Bootstrap

In earlier notes, we discussed several strategies for calculating the distribution of a statistic. We now introduce another technique for calculating the distribution of a statistic that is generally quite

useful and builds on the intuition of being able to accurately estimate the cumulative distribution function.

The distribution of statistic $Y = g(\mathcal{X})$ is a function of $g(\cdot)$ and the population distribution F_X . The typical issue, as we have seen above, is that we do not know F_X . One idea may be to approximate F_X by some close distribution. If statistic Y depends on the distribution of the data in a continuous enough fashion, we may get a good enough approximation. For example, in the i.i.d case $F_X(x) = \prod_{i=1}^n F(x_i)$, all we need is to approximate F , the cdf of one observation.

As one example, the Glivenko-Cantelli theorem above suggests that the empirical distribution $\hat{F}_n(x)$ may be a good approximation to $F(x)$. So, the idea is to run the same algorithm as one does with Monte-Carlo simulations, but simulate X_{ib}^* from distribution $\hat{F}_n(x)$ rather than from an unknown F . This specific procedure is called the *non-parametric bootstrap*:

- For $b = 1, \dots, B$, simulate $\mathcal{X}_b^* = (X_{1b}^*, \dots, X_{nb}^*)$, where the X_{ib}^* are drawn independently and uniformly with replacement from the set of initial observations $\{X_i, i = 1, \dots, n\}$ (each X_i has the same probability to be drawn);
- Calculate $Y_b^* = g(\mathcal{X}_b^*)$, e.g., the sample mean $\frac{1}{n} \sum_{i=1}^n X_{ib}^*$;
- If you are interested in:
 - the cdf of Y , then $F_Y(s) \approx \frac{1}{B} \sum_{b=1}^B I\{Y_b^* \leq s\}$;
 - probability Y gets into the set A : $P\{Y \in A\} \approx \frac{1}{B} \sum_{b=1}^B I\{Y_b^* \in A\}$.
 - α -quantile of Y : $q_Y(\alpha) \approx Y_{([\alpha B])}^*$, here $(\cdot)$ stands for the order statistics.
 - mean of Y : $\mathbb{E}Y \approx \frac{1}{B} \sum_{b=1}^B Y_b^*$.
 - variance of Y : $V(Y) \approx \frac{1}{B-1} \sum_{b=1}^B \left(Y_b^* - \frac{1}{B} \sum_{s=1}^B Y_s^* \right)^2$.

Larger B will help eliminate some part of approximation error (simulation error), but some part (due to approximating F) cannot be controlled. We will see many other ways to approximate $F(x)$, which would lead to a number of different bootstrap procedures.

There are often two ways to justify using the bootstrap. Typically we wish to claim that the distance between the true distribution of the statistic F_Y and the bootstrapped one F_{Y^*} is converging to zero in some sense as the sample size increases to infinity. One way is to show that F_Y is continuous in F in some sense (we have to be accurate as there are different metrics that can be introduced on the space of these distributions) and show that whatever approximation $\hat{F}$ we use, it converges to F in the proper metrics. The other way is to show that the distribution of Y converges somewhere (allow for asymptotic approximation) and that Y^* converges to the same distribution. We will discuss this many times in what follows.

The bootstrap is also useful for building intuition about what exactly the thought experiment behind our inferences is. Under random sampling, the thought experiment is that there is some large population and we draw our sample randomly from it; the source of uncertainty is who enters our sample. The bootstrap treats the sample we have as if it were the population and randomly samples from it — thus mimicking the thought experiment itself.

4 When the Bootstrap Fails

Both strategies for justifying the bootstrap rest on the same requirement: the distribution of the statistic must depend on the underlying distribution in a sufficiently continuous way, so that a good approximation to F translates into a good approximation to F_Y . There is a well-known list of settings in which this requirement breaks down, and in which the bootstrap fails alongside standard asymptotic approximations:

- Weak identification: When a parameter is only weakly identified, the sampling distribution of standard statistics depends on the strength of identification, a feature of the population that cannot be consistently estimated. The mapping from the population distribution to the sampling distribution is discontinuous at the point of non-identification, so resampling from $\hat{F}_n$ need not reproduce the correct distribution.
- Moment inequalities and parameters on the boundary: When the parameter lies on (or near) the boundary of the parameter space, or is restricted only by moment inequalities, the limiting distribution of the statistic changes discontinuously depending on which constraints bind. The bootstrap must in effect learn from the data whether each constraint binds, and it cannot do so accurately enough to reproduce the correct limit.
- Post-model-selection inference: When a statistic is computed after a data-driven model-selection step, its sampling distribution reflects the randomness of the selection itself and changes discontinuously in the parameters that govern which model is selected. The bootstrap replicates the noise in the selection step but not the correct limiting distribution.
- Non-smooth and extremum settings: When the statistic is not a smooth functional of the distribution of the data, such as a sample maximum or an estimator defined by a non-smooth objective function, the bootstrap distribution can converge to the wrong limit. For instance, a bootstrap draw can never exceed the observed sample maximum, so the bootstrap distribution of the maximum piles up on the top few order statistics rather than mimicking the true sampling distribution.

Some of these settings will reappear in later courses in the first-year econometrics sequence and particularly in second-year econometrics courses.